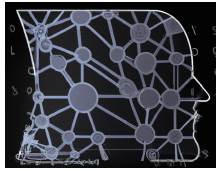
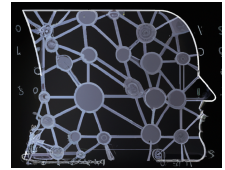
# Applied Machine Learning

# Time Series



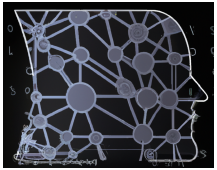
**Learning goals**

- What is a time series?
- Important definitions
- Feature engineering for time series
- Time series cross-validation



# Applied Machine Learning

# RUNNING EXAMPLE: THE BIKE SHARING DATASET



Hourly bike rental data from Washington D.C.

| | holiday | workingday | weather | temp | feel_temp | humidity | windspeed | datetime | count |
|---|---|---|---|---|---|---|---|---|---|
| 0 | False | False | clear | 9.84 | 14.395 | 0.81 | 0.00 | 2011-01-01 00:00:00 | 16 |
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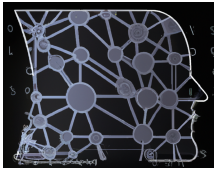

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Task: Predict hourly rental count

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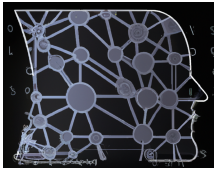

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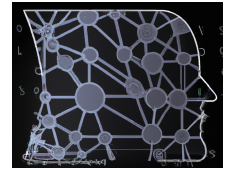
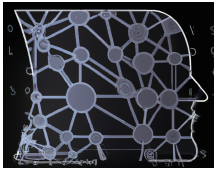
Task: Predict hourly rental count Question: how to feed this to a standard ML model?

# KEY DEFINITIONS



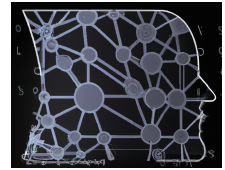
- **Time series** $\{y_t\}_{t=1}^{T}$: ordered observations indexed by time:
  - Timestamps - Specific instants in time.
  - Fixed periods - Such as the whole month of January 2017, or the whole year 2020.
  - Intervals of time - Indicated by a start and end timestamp. Periods can be thought of as special cases of intervals.
  - Experiment or elapsed time - Each timestamp is a measure of time relative to a particular start time (e.g., the diameter of a cookie baking each second since being placed in the oven), starting from 0.
- **Exogenous variable** / **covariate**: external feature $x_t^{(j)}$ influencing $y_t$
- **Trend**: long-term increase or decrease
- **Seasonality**: systematic, calendar-related patterns
- **Stationarity** (briefly): distribution of $y_t$ does not change over time

# TASKS



- Time Series Analysis Decompose time series into trends for understanding $\rightarrow$ time series course
- Time Series Forecasting
  Fit a model on historical data and make predictions about the future.
- Time Series Classification
  Make predictions about one complete time series, e.g., one recording of an experiment.

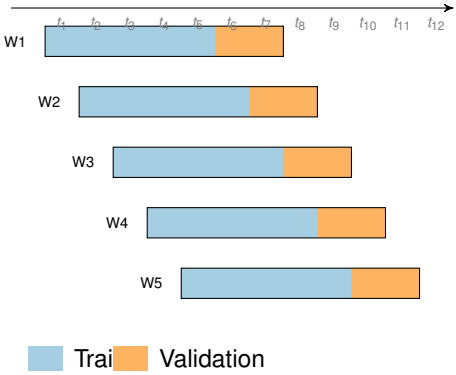
# HOW TO TACKLE TIME SERIES?

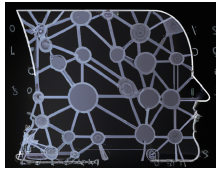


- Statistical Models ARIMA (Autoregressive integrated moving average) and ETS (Error, Trend, Seasonality) $\rightarrow$ time series course
- Recurrent Models
  Deep learning architectures that track the state of data over time $\rightarrow$ part of the deep learning lecture
- Transformer Models
  Deep learning architecture that can treat series data in an auto-regressive manner $\rightarrow$ part of the deep learning lecture
- Feature Engineering for Machine Learning
  Make predictions about one complete time series, e.g., one recording of an experiment.

# Time-Series Cross-Validation

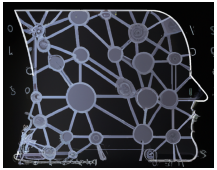
# CHRONOLOGY-AWARE SPLITS



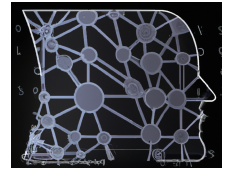
- **Expanding window**: grow train set, roll validation forward
- **Sliding window**: fixed-width train and validation segments
- Validation window should match the application forecasting windows

# AVOIDING DATA LEAKAGE



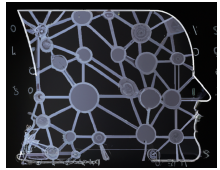
- Compute lagged / rolling features *after* defining the split or in a way it takes the split into account
- No peeking into future data when standardizing / scaling
- Requires special cross-validation methods

# EVALUATION METRICS



- **MAE** $= \frac{1}{n} \sum |y_t - \hat{y}_t|$ (robust to outliers)
- **RMSE** (penalizes large errors)
- **SMAPE**: symmetric MAPE avoids division issues when $y_t$ near 0
  - Symmetric Mean Absolute Percentage Error
  - SMAPE $= \frac{100}{n} \sum_{t=1}^{n} \frac{|F_t - A_t|}{|A_t| + |F_t|}$

# SIMPLE BASELINE



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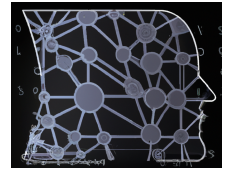
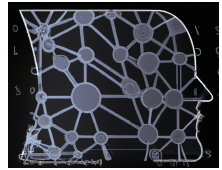
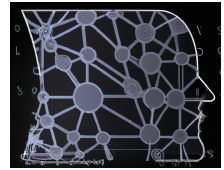


$$y_{t+1} = y_t$$



# Feature Engineering

# CALENDAR FEATURES



- Day-of-Week, Day-of-Month, Week Number, Month, Quarter
- Boolean flags: public holidays, weekend, promotion days, end-of-month

**Why?**

Capture human or process-driven periodicity.

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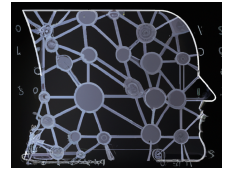


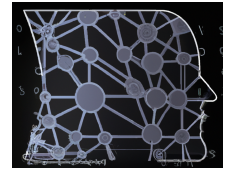
- Day-of-Week, Day-of-Month, Week Number, Month, Quarter
- Boolean flags: public holidays, weekend, promotion days, end-of-month

**Why?**

Capture human or process-driven periodicity.

Caveat: pay attention to the calendar if the events do not follow the Gregorian calendar

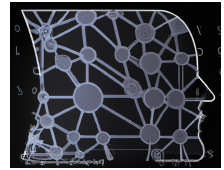
# CYCLIC ENCODING

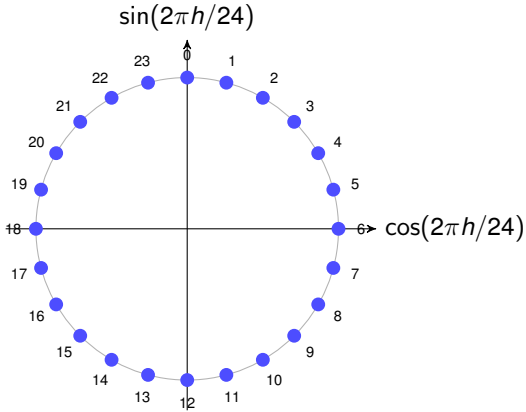


$$\sin(2\pi \frac{hour}{24}),\ \cos(2\pi \frac{hour}{24})$$

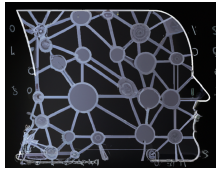
- Converts periodic integers (hour 0–23, month 1–12) into continuous space
- Eliminates artificial jump between $23 \rightarrow 0$

# CYCLIC ENCODING VISUALIZATION (HOUR 0–23)





# LAGGED & AUTO-REGRESSIVE FEATURES



- Plain lags: $y_{t-1},\ y_{t-7},\ y_{t-14}$
- Multi-step: include future-known covariates (e.g., planned price)
- Combine with calendar features for interactions

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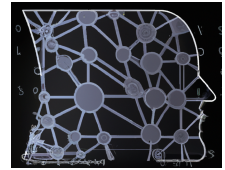

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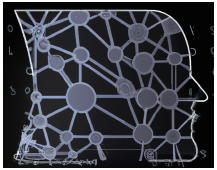
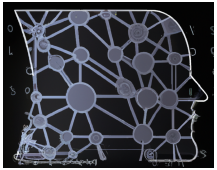
| | datetime | count | lag_168 | lag_336 | lag_504 | lag_672 |
|---|---|---|---|---|---|---|
| 17374 | 2012-12-31 19:00:00 | 119 | 18 | 507 | 564 | 692 |
| 17375 | 2012-12-31 20:00:00 | 89 | 23 | 340 | 427 | 471 |
| 17376 | 2012-12-31 21:00:00 | 90 | 22 | 200 | 300 | 300 |
| 17377 | 2012-12-31 22:00:00 | 61 | 12 | 120 | 245 | 221 |
| 17378 | 2012-12-31 23:00:00 | 49 | 11 | 54 | 126 | 144 |

# ROLLING / EXPANDING STATISTICS



- Rolling mean / std / min / max over window
- Expanding mean: trend indicator
- Percentiles: 25th, 75th for distribution shape

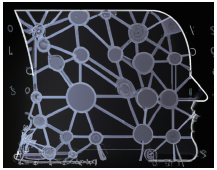
# INTERACTION FEATURES



- Lag $\times$ Holiday flag
- Weather $\times$ Weekend indicator
- Captures non-linear, conditional effects

# Wrap-Up

# KEY TAKEAWAYS



- Feature engineering often outweighs model complexity in time-series ML.
- Use chronology-aware CV to obtain honest performance estimates.
- Always benchmark against a naive or seasonal naive baseline.